VOICE INPUT SIMULATOR FOR KEYBOARD

Submitted By

MUHAMMAD EHTSHAM
(Regd. No. 2023-uok-03922)

HAMMAD SABIR
(Regd. No. 2023-uok-03914)

SANA ULLAH
(Regd. No. 2023-uok-03920)

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University of Kotli Azad Jammu and Kashmir
Faculty of Computing & Engineering
Department of CS&IT

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A Report

Submitted in partial fulfillment of the requirements for the 5th semester project of

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University of Kotli Azad Jammu and Kashmir
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1. INTRODUCTION

In modern computing systems, user input plays a very important role in human-computer interaction. Traditionally, keyboards have been the primary input device used to enter data and commands into a system. However, with advancements in technology, voice recognition has emerged as a powerful alternative method of interaction that enhances accessibility and ease of use.

This project, **"Keyboard Software with Voice Capture"**, focuses on combining both keyboard and voice input methods into a single system. The purpose of this system is to simulate how a keyboard device driver works while also integrating voice input functionality. A device driver acts as a bridge between hardware devices (such as keyboard and microphone) and the operating system, allowing smooth communication and data transfer.

2. PROJECT OBJECTIVE

The objective of this project is to design and simulate a Keyboard Device Driver that captures keyboard inputs and integrates voice input functionality.
The system aims to demonstrate how a device driver communicates between hardware (keyboard and microphone) and the operating system, allowing users to input commands using both keyboard typing and voice recognition**.**

2.1 Problem Statement:

Traditional input systems rely only on keyboard input. This project solves the problem by simulating a device driver that supports dual input modes (keyboard + voice) for improved accessibility and user interaction.

3. FUNCTIONAL REQUIREMENT

The system shall provide the following functionalities:

3.1 Voice and Keyboard Input Processing

- The system shall accept user input through both keyboard and voice commands.
- The system shall allow users to switch between keyboard input mode and voice input mode**.**
- The system shall remain active and continuously accept input until manually stopped.

-

Figure 1.1 Voice and Keyboard Input Processing

3.2 Keyboard Input Handling

- The system shall capture each key pressed by the user.
- The system shall display captured key presses on the screen in real time.
- The system shall simulate basic keyboard device driver behavior to demonstrate low-level input handling.

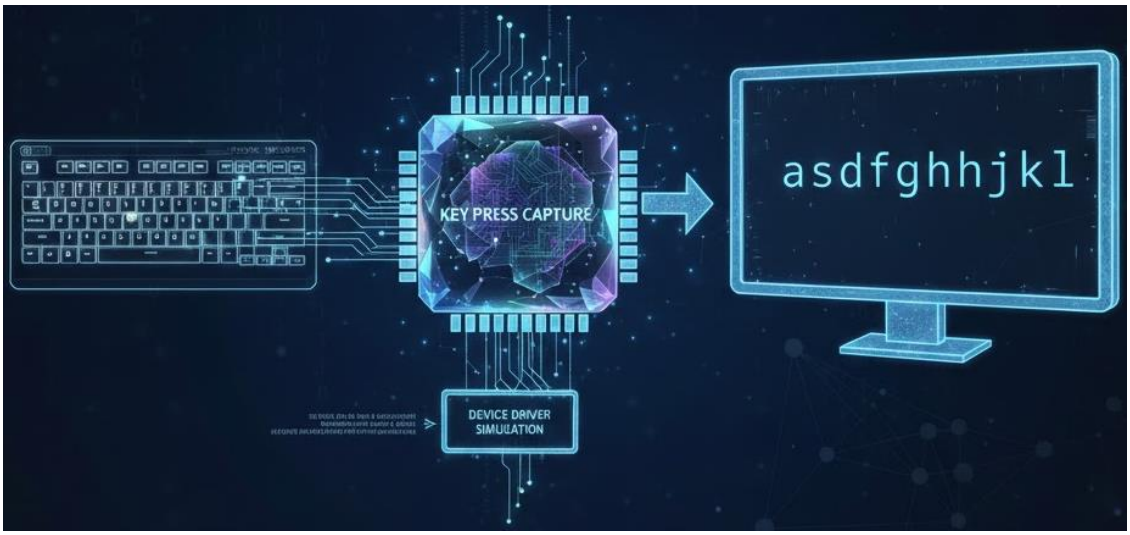

Figure 1.2 Keyboard Input Handling

3.3 Voice Input Capture

- The system shall record voice input using a connected microphone.
- The system shall accept voice commands in English and Urdu.

- The system shall continuously listen for voice commands after activation.

Figure 1.3 Voice input capture

3.4 Speech-to-Text Conversion

- The system shall convert recorded voice input into text using speech recognition technology.
- The converted text shall be displayed on the screen instantly.

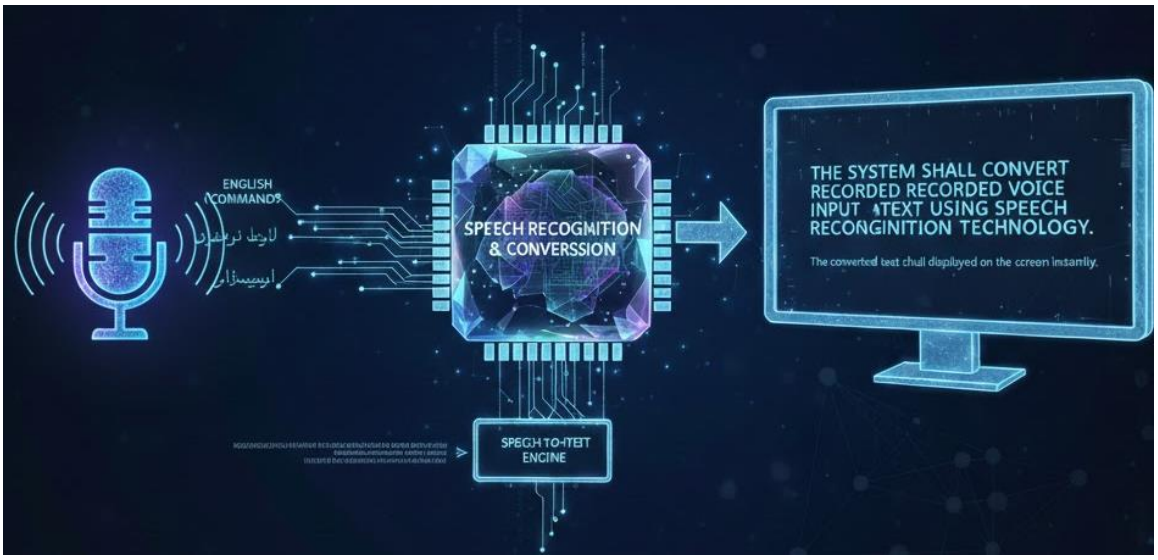

Figure 1.4 Speech-to-Text Conversion

3.5 Text Translation

- The system shall translate input text into a selected target language.
- The system shall support multiple languages including Urdu, English, Arabic, Chinese, Hindi, and French etc**.**
- The translated text shall be displayed in real time.

3.6 Plug and Play Device Support

- The system shall automatically detect input devices (e.g., keyboard, microphone) when they are connected.
- The system shall not require manual configuration for newly connected devices.
- The system shall notify the user when a device is successfully connected or disconnected.

3.7 Intelligent System Decision Making

- The system shall analyze system conditions before executing commands.
- The system shall check internet availability before executing internet-dependent commands.
- The system shall explain failures or limitations verbally (e.g., no internet, app not found).

2.9 Error Handling

- The system shall detect invalid, unclear, or failed voice commands.
- The system shall display appropriate error messages.
- The system shall suggest possible solutions using voice feedback.

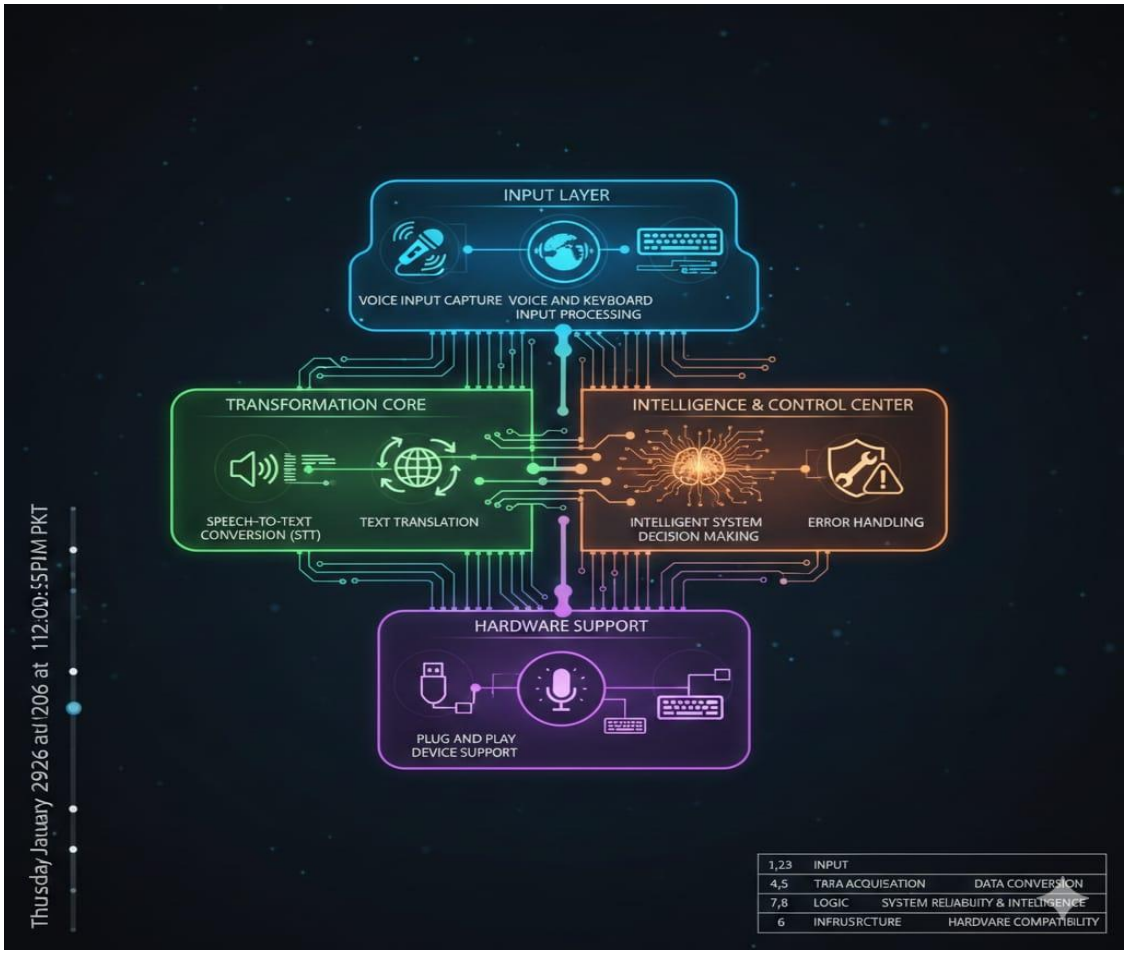

Figure 1.5

4. NON FUNCTIONAL REQUIREMENTS

- The system shall provide a user-friendly and interactive interface for easy use.
- The system shall respond quickly to keyboard and voice inputs.
- The system shall work on both Windows and Linux operating systems.
- The system shall accurately recognize common voice commands.
- The system shall support multiple languages for input and translation.
- The system shall remain stable during continuous use without crashing.
- The system shall securely process voice data and prevent unauthorized access.

5 INPUT REQUIREMENTS

The user will provide the following inputs:

- Keyboard inputs (alphabets, numbers, special characters)
- Voice input through microphone
- Control commands (start/stop voice capture, clear input, exit system)

6. OUTPUT REQUIREMENTS

The system will produce the following outputs:

- Display typed keyboard characters on the screen
- Display converted text from voice input
- Real-time input log
- Status messages (e.g., “Voice Capture Started”, “Input Recorded”)
- Error messages for invalid or unclear voice input

7. ASSUMPTIONS & CONSTRAINTS

Assumptions

- Microphone is available and properly configured
- Users speak clearly for accurate voice recognition
- All inputs are valid and meaningful

Constraints

- Voice recognition accuracy depends on background noise
- Internet may be required if using online speech recognition APIs
- System is limited to basic keyboard and voice commands
- Simulation only (not an actual kernel-level driver)

8. CONCLUSION

This project demonstrates a simulated keyboard device driver with voice input support. It allows users to input commands using both keyboard and voice, improving accessibility and user interaction. The system captures keyboard input, converts voice into text, supports multiple languages, and provides real-time feedback.

Although the system is only a simulation and not a real kernel-level driver, it effectively explains the basic concepts of device drivers and input handling. The project has some limitations such as internet dependency and background noise, but it provides a strong foundation for future improvements like offline voice recognition and advanced features.

9. REFERENCES

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